The ED Micro-Event Operator:

A Geometric Mechanism Classifier for Architectural Dynamics

Abstract

The micro-event operator is the diagnostic core of Event Density (ED), providing the mechanism-level insight required to understand and classify collapse behavior in architectural dynamical systems. While ED's law surfaces predict collapse time from initial geometry, they do not reveal the structural cause of collapse. The micro-event operator fills this gap by detecting the earliest geometric degeneracy—defined by the minimum PBC-corrected distance—and extracting a complete snapshot of the ring's geometry, dynamics, and mechanism at that moment. Using only geometric quantities such as $d_{\min(t)}$, circumradius trends, angular structure, and boundary proximity, the operator assigns one of five mechanism labels: inward-collapse, outward-PBC, PBC-corner, other-late, or DECAY. These mechanisms correspond to distinct regions of the ED manifold and define the boundaries between dynamical regimes. Through extensive validation across the ED-Arch scenarios, the operator demonstrates determinism, compositionality, scale-invariance, and robustness to perturbations. It is the tool through which the seven invariants of ED-Arch-21 are discovered and verified, revealing that ED is governed by architectural laws rather than emergent numerical artifacts. This paper formalizes the operator, its geometric foundations, its classification rules, and its role in mapping the structure of the ED manifold.

1. Introduction

Event Density (ED) requires more than a way to measure collapse time; it requires a way to understand why a collapse occurs. The ED micro-event operator provides this mechanism-level insight. It identifies the earliest moment when a ring's geometry becomes degenerate, extracts the geometric signature of that moment, and classifies the mechanism responsible. This transforms ED from a simulation into a diagnostic architecture.

Collapse time alone is a scalar quantity. It tells us when a collapse occurs, but not how or why. Mechanisms matter: inward collapses, outward PBC collapses, tangent-drift collapses, and DECAY trajectories all occupy distinct regions of the ED manifold. The micro-event operator reveals these regions by detecting the structural features that drive each collapse.

This paper formalizes the operator, its geometric foundations, and its classification rules. It shows how the operator identifies regime boundaries, supports compositionality analysis, and enables the extraction of the seven invariants that close the ED-Arch series. The operator is the methodological core of ED: the lens through which architectural dynamics become legible.

2. Background and Context

2.1 The ED Ontology (Brief Recap)

ED begins with a minimal ontology: rings of N particles arranged on a circle of diameter d, evolving ballistically under one of three gamma gates (inward, tangent, outward). Geometry is the primary driver of behavior. Two quantities are central:

- $d_{\min(t)}$: the minimum PBC-corrected distance between any two particles
- $R(t)$: the circumradius of the ring

Collapse occurs when $d_{\min(t)}$ falls below an architectural threshold. The law surfaces predict the collapse time χ, but they do not explain the mechanism. That is the role of the micro-event operator.

2.2 Why Mechanisms Matter

Two rings may share the same collapse time but follow entirely different dynamical paths. For example:

- An inward collapse is driven by radial contraction.
- An outward collapse is driven by PBC re-entry.
- A tangent-drift collapse is driven by geometric asymmetry.
- A DECAY trajectory never collapses at all.

These differences are structural. They define regime boundaries, determine how rings interact, and reveal the architecture of the ED manifold. Without mechanism classification, ED would be unable to distinguish these regimes or extract the invariants that govern them.

2.3 Role in the ED-Arch Series

The micro-event operator emerged gradually across the ED-Arch program:

- ED-Arch-13 introduced the first mechanism detector.
- ED-Arch-16 refined the classification rules and added PBC-corner detection.
- ED-Arch-21 used the operator to extract the seven invariants that close the theory.

By the end of the series, the operator had become indispensable. It is the tool that reveals the structure of the ED manifold and the laws that govern it.

3. Formal Definition of a Micro-Event

3.1 Collapse Condition

A micro-event occurs at the earliest simulation step t such that:

$$d_{\min(t)} \leq \text{MERGE_THR}$$

This is the architectural definition of collapse. It is independent of numerical parameters, integration schemes, or implementation details. It depends only on geometry.

3.2 The Micro-Event Snapshot

At the moment of collapse, the operator records a complete geometric and dynamical snapshot:

- collapse step t_c
- empirical collapse time χ_{emp}
- circumradius $R(t_c)$
- ring constant K
- full particle positions and velocities
- active gamma gate
- mechanism label

This snapshot is the atomic unit of ED dynamics. It captures the minimal information needed to classify the mechanism and place the event on the ED manifold.

3.3 Why Earliest Matters

The operator always selects the earliest collapse event. This ensures:

- mechanism purity — the geometry has not yet been distorted by late-stage interactions
- boundary clarity — regime transitions occur at the earliest degeneracy
- invariant extraction — the operator sees the structural cause, not the aftermath

Late-stage geometry can obscure the mechanism. Early-stage geometry reveals it.
This is why the micro-event operator is defined at the earliest collapse step.

4. Geometric Foundations

The micro-event operator is grounded entirely in geometry. It does not rely on forces, potentials, or continuous dynamics. Instead, it extracts structural information from the ring's geometry at the moment of collapse. Three geometric quantities form the foundation of the operator: the minimum PBC-corrected distance, the circumradius, and the angular structure of the ring.

4.1 Minimum PBC Distance

The primary geometric signal is the minimum PBC-corrected distance:

$$d_{\min}(t) = \min_{i \neq j} \|x_i(t) - x_j(t)\|_{\mathrm{PBC}}$$

This quantity detects the earliest degeneracy in the ring's geometry. Because PBC modifies the effective distance between particles, $d_{\min(t)}$ captures both local contraction and boundary-driven interactions. The collapse condition is defined entirely in terms of this quantity.

4.2 Circumradius and Radial Trends

The circumradius,

$$R(t) = \frac{1}{N} \sum_{i=1}^{N} \|x_i(t) - c(t)\|,$$

reveals whether the ring is contracting, drifting, or expanding. The sign of $\Delta R(t)$ is one of the strongest indicators of mechanism:

- $\Delta R < 0$: inward contraction
- $\Delta R \approx 0$: tangent drift
- $\Delta R > 0$: outward expansion

The operator uses these trends to distinguish inward, tangent, and outward regimes before collapse occurs.

4.3 Angular Structure

The angular positions of particles around the ring encode drift and DECAY behavior. Tangent rings exhibit slow, predictable angular drift, while DECAY trajectories maintain a stable angular configuration with no contraction. Angular structure is therefore essential for distinguishing tangent-drift collapses from DECAY.

4.4 Boundary Geometry

Periodic boundary conditions introduce additional geometric structure. When a particle approaches a domain edge or corner, the PBC-corrected distance may decrease even if the ring is expanding. This produces distinct collapse signatures:

- PBC-corner: collapse triggered by a particle near a domain corner
- Outward-PBC: collapse triggered by re-entry after expansion

These signatures are detectable only through PBC-aware geometry.

5. Mechanism Classification

Mechanism classification is the core function of the micro-event operator. At the moment of collapse, the operator assigns one of five mechanism labels based on geometric criteria. These labels correspond to distinct regions of the ED manifold and define the boundaries between dynamical regimes.

5.1 Inward-Collapse

Signature:

- monotone decrease in $d_{\min(t)}$
- $\Delta R(t) < 0$
- collapse occurs without PBC interaction

This is the purest form of collapse and corresponds to the inward law surface.

5.2 Outward-PBC

Signature:

- $\Delta R(t) > 0$
- ring expands until interacting with the boundary
- collapse triggered by PBC re-entry

This mechanism corresponds to the outward law surface and depends on domain geometry.

5.3 PBC-Corner

Signature:

- a particle approaches a domain corner
- PBC-corrected distances decrease asymmetrically
- collapse occurs before full re-entry

This mechanism is distinct from outward-PBC and marks a separate boundary region.

5.4 Other-Late

Signature:

- collapse occurs after $\chi>3$
- no clear radial or boundary driver
- geometry becomes degenerate only at late times

These events are rare but structurally important for identifying the edges of the ED manifold.

5.5 DECAY

Signature:

- no collapse within the simulation horizon
- $\Delta R(t) \approx 0$
- stable angular configuration
- tangent-like drift with no contraction

DECAY is not a collapse mechanism but a stable sub-regime of the ED manifold.

5.6 Decision Tree

The operator uses a deterministic decision tree:

1. Check for collapse within horizon
   - if none → DECAY

2. Evaluate $\Delta R(t)$
   - negative → inward-collapse
   - positive → outward or PBC

3. Evaluate PBC proximity
   - near corner → PBC-corner
   - near edge → outward-PBC

4. Otherwise → other-late

This classification is architectural: it depends only on geometry, not on simulation details.

6. Regime Boundaries

The ED manifold is divided into distinct dynamical regimes, each governed by a law surface. The micro-event operator identifies the boundaries between these regimes by detecting changes in mechanism.

6.1 Inward ↔ Tangent Boundary

This boundary occurs where radial contraction transitions to tangent drift. It is detected when $\Delta R(t)$ changes sign and the mechanism switches from inward-collapse to tangent-drift behavior.

6.2 Tangent ↔ Outward Boundary

This boundary marks the transition from drift to expansion. It is detected when $\Delta R(t)$ becomes consistently positive and the ring begins to move toward the boundary.

6.3 Outward ↔ PBC Boundary

This boundary occurs when outward expansion meets the domain geometry. The operator detects this transition when PBC-corrected distances begin to decrease despite increasing $R(t)$.

6.4 DECAY Region

The DECAY region is a stable angular sub-regime where no collapse occurs. It is bounded by tangent-drift behavior on one side and outward expansion on the other. The operator identifies this region by the absence of collapse and the stability of angular structure.

6.5 Boundary Detection via Micro-Events

Because the operator records the earliest collapse event and its mechanism, it provides a clean, structural map of the ED manifold:

- mechanism labels define regime regions
- transitions between labels define boundaries
- repeated sampling across (d,K,γ) reconstructs the manifold

This is how ED-Arch-17 and ED-Arch-20 produced the chi manifold and gamma sheets.

7. Empirical Validation

The micro-event operator is not a theoretical construct alone; it is validated across the full suite of ED-Arch scenarios. These experiments demonstrate that the operator is deterministic, geometry-aware, and stable across ring sizes, diameters, gamma gates, and multi-body configurations. The operator's mechanism labels match the structural behavior observed in every regime of the ED manifold.

7.1 Scenario H: Golden Inward Collapse

Scenario H provides the canonical inward-collapse trajectory. The operator consistently identifies:

- monotone decrease in $d_{\min(t)}$
- negative $\Delta R(t)$
- no PBC interaction

The mechanism label is always inward-collapse, and the empirical χ_{emp} matches the inward law surface with zero deviation. This scenario establishes the operator's correctness in the simplest regime.

7.2 Triads: Three-Cluster Compositionality

In the triads scenario, three independent inward collapses occur simultaneously. The operator detects three micro-events, each with:

- identical mechanism labels
- identical χ_{emp} values
- independent geometric signatures

This confirms mechanism compositionality: the operator treats each ring independently, even when they coexist in the same field. This is a key requirement for INV-21-1.

7.3 Compositionality Matrix (3×3 Gamma Pairs)

The 3×3 matrix of gamma-pair interactions tests all combinations of inward, tangent, and outward gates. The operator correctly identifies:

- inward-collapse for inward–inward
- DECAY for tangent–tangent
- outward-PBC for outward–outward
- mixed mechanisms for cross-gamma pairs

The three DECAY cases correspond exactly to the tangent–tangent and tangent-outward regimes, confirming the stability of the DECAY sub-regime.

7.4 Scaling Across N=3…12

The operator is stable across ring sizes. For each N, the mechanism labels and χ_{emp} values match the predicted law surfaces. No misclassifications occur. This demonstrates that the operator is scale-invariant, a necessary property for architectural systems.

7.5 Robustness to Perturbations

Pause-resume experiments, center offsets, and small positional perturbations do not change the mechanism label. The operator detects the same micro-event with the same geometric signature. This robustness is essential for INV-21-2 (memorylessness) and INV-21-4 (pause-resume invariance).

8. Role in Invariant Extraction

The micro-event operator is the tool through which the invariants of ED-Arch-21 are discovered and verified. Each invariant corresponds to a structural regularity in the ED manifold, and each regularity is revealed through mechanism-level analysis.

8.1 INV-21-1: Compositionality

The operator shows that mechanism labels combine cleanly across bodies. In multi-ring systems, each ring's mechanism is independent of the others. This compositionality is visible only through micro-event snapshots.

8.2 INV-21-2: Memorylessness

Pause-resume experiments demonstrate that the mechanism depends only on the current geometry. The operator detects identical micro-events before and after the pause, confirming that ED dynamics evolve on a Markovian surface.

8.3 INV-21-3: Tangential Drift Law

The operator identifies tangent-drift micro-events and extracts the sideways drift $d_{\mathrm{sw,tangent}}$. This drift is invariant across perturbations and depends only on (d,K). Without mechanism classification, the drift law would be invisible.

8.4 INV-21-4: Pause-Resume Invariance

The operator detects identical micro-events before and after a pause, confirming that collapse time and mechanism are architectural, not dynamical artifacts.

8.5 INV-21-5: Perturbation Hardness Ordering

By comparing mechanism labels across perturbations, the operator reveals the stability ordering:

- inward = maximally stable
- tangent = neutrally stable
- outward = minimally stable

This ordering is invariant across all N and d.

8.6 INV-21-6: DECAY Angular Sub-Regime

The operator identifies DECAY trajectories by the absence of collapse and the stability of angular structure. This sub-regime is a structural feature of the ED manifold.

8.7 INV-21-7: Ghost Compositionality

When one ring collapses and another does not, the operator shows that the non-collapsing ring's mechanism and predicted behavior remain unchanged. Collapse does not propagate across bodies.

9. Implementation

The micro-event operator is implemented as a deterministic, inspectable component of the ED codebase. Its structure reflects the architectural principles of ED: minimalism, clarity, and reproducibility.

9.1 The MicroEvent Dataclass

The operator stores each micro-event in a structured dataclass containing:

- collapse step
- χ_{emp}
- $R(t_c)$
- K
- particle positions and velocities
- gamma gate
- mechanism label

This structure ensures that micro-events are self-contained and serializable.

9.2 The Operator Function

The operator is invoked during simulation with detect_events=True. It monitors $d_{\min(t)}$ at each step and triggers when the collapse condition is met. The mechanism classification is performed immediately, using only geometric information.

9.3 Integration with run_{ring}

The operator is tightly integrated with the run_{ring} function. This ensures:

- deterministic event detection

- consistent mechanism classification
- seamless integration with scenario runners

The operator does not modify the simulation; it observes it.

9.4 JSON Serialization

All micro-events are serialized to JSON for reproducibility. This enables:

- figure generation
- invariant extraction
- cross-scenario comparison
- long-term archival of results

The JSON format ensures that micro-events remain inspectable and portable.

10. Discussion

The micro-event operator transforms Event Density from a simulation into a scientific instrument. By identifying the earliest geometric degeneracy and classifying the mechanism responsible, the operator reveals the structural organization of the ED manifold. This mechanism-level insight is essential for understanding regime boundaries, extracting invariants, and demonstrating that ED is governed by architectural laws rather than emergent numerical artifacts.

Several themes emerge from the operator's behavior across the ED-Arch program.

10.1 Mechanisms as Architectural Objects

The operator shows that collapse mechanisms are not incidental outcomes of simulation dynamics; they are architectural objects. Inward-collapse, outward-PBC, tangent-drift, PBC-corner, and DECAY each correspond to distinct geometric signatures and occupy well-defined regions of the ED manifold. These mechanisms are reproducible, predictable, and invariant across ring sizes, diameters, and initial conditions.

10.2 Geometry as the Source of Predictive Power

The operator's classification rules depend only on geometry: $d_{\min(t)}$, $R(t)$, angular structure, and PBC proximity. This geometric foundation explains why ED's law surfaces are so accurate. Collapse time is not an emergent property of a complex dynamical system; it is a geometric function of initial conditions. The operator exposes this structure by detecting the geometric cause of collapse.

10.3 Regime Boundaries and the ED Manifold

The operator provides a direct method for identifying regime boundaries. Transitions between inward, tangent, and outward behavior correspond to changes in mechanism labels. PBC-corner events mark the edges of the outward regime. DECAY trajectories define a stable angular sub-regime. These boundaries are architectural: they arise from the structure of the ED ontology, not from simulation artifacts.

10.4 Compositionality and Multi-Body Dynamics

The operator demonstrates that mechanisms combine cleanly across bodies. In multi-ring systems, each ring's mechanism is independent of the others. This compositionality is essential for ED's scalability and for the invariants of ED-Arch-21. Without the operator, compositionality would be invisible.

10.5 Invariants as Mechanism-Level Laws

The operator is the tool through which the seven invariants are discovered and verified. Each invariant corresponds to a structural regularity in mechanism behavior. Memorylessness, pause-resume invariance, perturbation hardness ordering, and ghost compositionality all emerge from mechanism-level analysis. The operator is therefore not just a classifier; it is the gateway to ED's architectural laws.

10.6 ED as a Diagnostic Architecture

With the micro-event operator, ED becomes:

- inspectable — mechanisms are visible and interpretable
- falsifiable — mechanism predictions can be tested
- extensible — new bodies and interactions can be added
- architectural — governed by structure, not simulation details

The operator is the methodological core of ED. It reveals the architecture underlying micro-dynamics and provides the foundation for the invariants that close the ED-Arch series.

11. Conclusion

The micro-event operator is the methodological core of Event Density. It provides the mechanism-level insight required to understand, classify, and predict collapse behavior in architectural dynamical systems. By detecting the earliest geometric degeneracy and assigning a mechanism label based solely on structural features, the operator reveals the organization of the ED manifold and the boundaries between its dynamical regimes.

The operator's geometric foundation—minimum PBC-corrected distance, circumradius trends, angular structure, and boundary proximity—ensures that mechanism classification is deterministic, reproducible, and independent of simulation details. This architectural clarity is what allows ED to produce accurate law surfaces, stable regime boundaries, and a coherent manifold structure across all ring sizes, diameters, and gamma gates.

Through extensive validation across the ED-Arch scenarios, the operator demonstrates compositionality, scale-invariance, and robustness to perturbations. These properties are essential for the extraction of the seven invariants that close the ED-Arch series. Without the operator, the invariants would be invisible; with it, they become structural laws.

The micro-event operator transforms ED from a numerical experiment into a scientific instrument. It makes mechanisms legible, boundaries detectable, and invariants discoverable. It is the lens through which the architecture of micro-dynamics becomes visible. As ED expands to multi-body universes, higher-order interactions, and new architectural regimes, the operator will remain the foundational tool for understanding the structure of collapse.